

Department of Software Engineering

Temperature & Humidity Monitoring System

*Real-Time Temperature and Humidity Monitoring using Arduino UNO, DHT11,
and I2C LCD*

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Course

Digital Logic Design Lab

Department

Department of Software Engineering

Project Code

```
//Humidity sensor
//scientist BENIELS LAB

#include <DHT.h>
#include <LiquidCrystal_I2C.h>
#include <Wire.h>

LiquidCrystal_I2C lcd(0x27,16,2); // set the LCD address to 0x27 for a 16 chars
and 2 line display

//Constants
#define DHTPIN 2    // what pin we're connected to
#define DHTTYPE DHT11 // DHT 11
DHT dht(DHTPIN, DHTTYPE); //// Initialize DHT sensor for normal 16mhz Arduino

//Variables
//int chk;
int h; //Stores humidity value
int t; //Stores temperature value

void setup()
{
  Serial.begin(9600);
  Serial.println("Temperature and Humidity Sensor Test");
  dht.begin();
  lcd.init(); //initialize the lcd
  lcd.backlight(); //open the backlight
}

void loop()
{
  //Read data and store it to variables h (humidity) and t (temperature)
  // Reading temperature or humidity takes about 250 milliseconds!
  h = dht.readHumidity();
  t = dht.readTemperature();

  //Print temp and humidity values to serial monitor
  Serial.print("Humidity: ");
  Serial.print(h);
  Serial.print(" %, Temp: ");
  Serial.print(t);
  Serial.println(" ° Celsius");

  // set the cursor to (0,0):
```

```
// print from 0 to 9:
```

```
lcd.setCursor(0,0);  
lcd.print("Temperature:");  
lcd.print(t);  
lcd.print("C");
```

```
lcd.setCursor(1, 1);  
lcd.print("Humidity:");  
lcd.print(h);  
lcd.print("%");
```

```
delay(1000); //Delay 1 sec.  
}
```